

Mohin Uddin

SKILLS

Programming	Python, SQL, C++, R, VBA, MATLAB, LaTeX
Data	Snowflake, PostgreSQL, MSSQL, dbt, Alteryx
ML	Scikit-learn, TensorFlow, XGBoost, LightGBM, Optuna, Statsmodels, Pandas, Polars, NumPy, GeoPandas
AI Tooling	Claude Code, Claude Design, Cursor, Granola
Cloud/MLOps	AWS (SageMaker, Athena, S3, ECR), Azure DevOps, Docker, Kubernetes, Kubeflow, Airflow, ArgoCD, Buildkite, GitHub, Datadog
Visualisation	Looker, Power BI, TIBCO Spotfire, Pyramid Analytics
Insurance	Radar, Emblem, Akur8, Tyche, Renew
Project Management	Jira, Confluence, Notion, Whimsical
Languages	English (native), Bengali (native)

EDUCATION

Durham University - MSc Mathematical Sciences	OCT 2018 - SEP 2019
Lancaster University - BSc (Hons) Mathematics	OCT 2015 - JUL 2018

PROFESSIONAL EXPERIENCE

Zego - Lead Data Scientist **FEB 2025 - PRESENT**

- Championed and gained CEO/COO approval for Zego's first-ever production gradient boosting machine (GBM), building the end-to-end modelling and deployment framework from scratch.
- Drove adoption of GBMs across pricing and Data Science, developing models directly and providing technical leadership and review for more than 10 production deployments.
- Built and productionised Zego's first market clustering model within the pricing function, using chained gradient boosting machines to generate features for segmentation.
- Built the production ML framework for new models now used across pricing and data science, standardising packaging, CI/CD, containerisation, and GitOps in collaboration with MLE and Systems Engineering.
- Leading the operationalisation of ML across the pricing function, including SageMaker implementation, automated retraining pipelines, model monitoring, and model performance comparison frameworks.
- Developed a Python integration library for Akur8 and refreshed the PCI GLM model to improve responsiveness to market pricing movements.
- Proactively identified and escalated material data quality, pricing, and operational risks across the business, including aggregator discrepancies, blocked vehicle logic, and fraudulent policy detection.
- Acted as Data Science lead for the SageMaker proof of concept, representing the DS function in collaboration with MLE leadership and advocating for wider adoption across teams.
- Served as interim manager for a junior data scientist following team restructuring, while continuing to lead onboarding support and code reviews across senior members (heads of DS) and junior team members.
- Regularly engaged with SLT including the COO, directly influencing pricing decisions — for example, identifying that sense scores were not being applied to new business policies, which was escalated and actioned the same day.
- Conducted 60+ interviews and technical assessments across DS, MLE, and actuarial hiring, helping shape the technical interview process across teams.
- Represented the Data Science function at company onboarding events, presenting to 30+ new joiners.

Marshmallow - Senior Pricing Data Scientist **SEP 2023 - FEB 2025**

- Built 5 targeted GBMs (including Migrant Diversification and Expansion Groups), owning all development and managing a junior across the project. Initially deployed at 20% of weekly new business, scaling to 70% due to strong performance, generating over £130m in written premium for 2024.
- Led the development of the company's first burning-cost GBM from scratch, including model build, evaluation, and delegating workstreams to junior team members.
- Fully automated the model data extraction, modelling, and validation pipeline for aggregator data. Also automated tenure bias checks that were previously performed manually.
- Managed the Renewal Kitchen Sink project through a direct report, testing elasticity-based pricing on renewals using a market model.
- Designed and executed an A/B test on direct customers using a market model to assess price elasticity, analysing results to inform pricing strategy.

- Identified elevated TPPD frequency and performed segmental loss ratio analysis, presenting findings to the loss ratio forum including the CEO, leading to adjustments in the live pricing algorithm.
- Regularly presented to SLT including the CEO. Served as net pricing's representative to the acquisition tribe, coordinating the team's actions based on tribe requests.
- Set up AWS SageMaker Notebook Instances for the team, liaising with Machine Learning Engineering to enable cloud computing in the pricing workflow.
- Conducted over 20 interviews leading to hiring decisions across the pricing team, and developed a comprehensive onboarding guide after identifying inefficiencies in the existing process.

Direct Line Group - P&U Senior Data Scientist

JUL 2022 - SEP 2023

- Built the highest value-adding model of 2022 (Market Premium GBM) using aggregator data, deployed into production and used for price optimisation.
- Developed GLMs and GBMs using Radar and Emblem across personal lines.
- Presented machine learning models to key stakeholders, securing approval for multiple models through the model governance board.
- Mentored junior colleagues through recorded Python coaching sessions, code reviews, and ad-hoc programming support.

AEGIS London - Actuarial Data Scientist

JUN 2021 - JUN 2022

- Sole data scientist in the company, operating across reserving, capital, and pricing - leading and presenting all data science projects.
- Built multivariate linear regression, random forest, and gradient boosting models across multiple lines of business, developing an automated risk grader.
- Created a reusable library of machine learning algorithm templates in R and Python with example datasets, enabling the wider team to adopt data science techniques.
- Delivered ad-hoc analysis for key stakeholders across various lines of business, producing reports that directly influenced changes to rating factors.
- Built reserving and capital dashboards in Pyramid Analytics, improving data visibility for stakeholders.
- Played a key role in the decision to migrate the company's pricing platform to hx Renew.
- Priced new and renewal policies with various layers on request from underwriters.

Autoprotect - Pricing Analyst

JAN 2021 - APR 2021

- Led the migration of pricing data from MSSMS to AWS (Athena, S3 & SageMaker), enabling cloud-based querying and analysis for the team.
- Performed risk analysis across multiple rating variables on GAPRTI motor policies, informing pricing decisions.
- Built a RAG report for all GAPRTI policies to calculate the ULR, automating PDF generation from Excel using VBA macros.

EY (Ernst & Young) - Actuarial Consultant

SEP 2019 - OCT 2020

- Built a model to estimate Technical Provisions for a Lloyd's of London syndicate, one of the largest by GWP and market capitalisation, validating the reasonableness of reserves.
- Projected triangle data in WinRes using Cape Cod, IBF, ICL, and ELR methods to support reserve opinions.
- Managed client relationships across all projects, providing regular progress updates and coordinating data requirements.
- Contributed to the production of Statements of Actuarial Opinion (SAOs) and GAAP ASRMs for audit purposes.

EXTRACURRICULAR ACTIVITIES

Cutting-Edge AI: Deep Reinforcement Learning in Python

SEP 2024

- Covered deep reinforcement learning techniques including evolution strategies, A2C, and DDPG.

Quantitative Finance & Algorithmic Trading in Python

APR 2023

- Covered financial engineering fundamentals including the Markowitz model, CAPM, and Black-Scholes model.

Institute and Faculty of Actuaries - Certificate in Data Science

SEP 2021 - DEC 2021

- Jointly awarded by the Southampton Data Science Academy and the IFoA, covering applied data science techniques within actuarial practice.

INTERESTS

- Royal Statistical Society member. Interests include building computers, exploring emerging AI tools and large language models, entrepreneurship, martial arts, gaming, and reading.